

Circles

Tangent to a circle, how many tangents can be drawn from a point, why the tangent is perpendicular to the radius, and why the two tangents from an external point are equal in length, with proofs and worked examples.

Introduction

Draw a circle and a straight line on the same plane. Only three things can happen: the line does not meet the circle at all, it meets it at exactly one point, or it meets it at two points. When it meets at two points we call it a **secant**. When it meets at exactly one point we call it a **tangent**. This chapter is only about tangents, and it contains just two theorems that the board asks again and again.

To talk about tangents clearly, it helps to think about where the point is with respect to the circle. A point can be **inside** the circle, **on** the circle, or **outside** the circle. How many tangents you can draw depends entirely on this.

Tangent to a Circle

A tangent to a circle is a line that touches the circle at exactly one point and does not cross into its interior at that point. That single point is called the **point of contact**.

Tangent and Point of Contact

A **tangent** to a circle is a line that intersects the circle at exactly one point. The point where the line touches the circle is called the **point of contact**.

A **secant**, by contrast, intersects the circle at two distinct points.

Imagine a secant cutting the circle at two points A and B. Now slowly rotate or shift the line so that A and B come closer together. The moment the two points coincide and become a single point, the secant becomes a tangent. In simple words, a

tangent is the limiting position of a secant when the two intersection points get infinitely close.

At the point of contact, the circle and the tangent share exactly one point. If the line went through the interior of the circle, it would have two points in common and would be a secant, not a tangent.

Number of Tangents from a Point

How many tangents can pass through a given point? The answer depends on where the point lies.

Number of Tangents from a Point

- 1. Point inside the circle:** no tangent can be drawn. Any line through an interior point must cut the circle at two points.
- 2. Point on the circle:** exactly one tangent can be drawn at that point.
- 3. Point outside the circle:** exactly two tangents can be drawn to the circle.

This is easy to picture. If you are standing inside a walled garden, you cannot draw a line that just touches the wall and passes through you, every line through you will cut the wall twice. If you stand exactly on the wall, there is just one line that touches the wall at your feet and goes along it. If you stand outside, you can reach the wall with your left hand and your right hand, two distinct lines that each just touch the wall.

When the point is outside, the two tangents go to two different points of contact on the circle. The segment between the external point and either point of contact is called the **length of the tangent** from that external point.

Worked Example

How many tangents can be drawn to a circle of radius 5 cm from a point P that is (i) 3 cm, (ii) 5 cm, and (iii) 8 cm from the centre?

- $3\text{ cm} < 5\text{ cm}$, P is inside the circle, so **no tangent** can be drawn.
- $5\text{ cm} = 5\text{ cm}$, P is on the circle, so **exactly one** tangent can be drawn.
- $8\text{ cm} > 5\text{ cm}$, P is outside the circle, so **two tangents** can be drawn.

Theorem: Tangent is Perpendicular to Radius

This is the first and most important theorem of the chapter. Every question about tangents uses it, because it lets you create a right angle where the tangent meets the radius.

Theorem 10.1, The tangent at any point of a circle is perpendicular to the radius through the point of contact.

If a line touches a circle at point P and O is the centre, then $OP \perp$ tangent at P.

Proof. Let AB be the tangent to the circle with centre O at the point of contact P. We want to show that OP is perpendicular to AB.

Take any other point Q on the line AB, different from P. Since AB touches the circle at only one point P, every other point Q on AB must lie outside the circle. So OQ is greater than the radius OP, that is, $OQ > OP$.

This is true for every point Q on AB other than P. So among all segments joining O to a point on AB, OP is the shortest. And we know from geometry that the shortest distance from a point to a line is the perpendicular distance. So OP must be perpendicular to AB.

$OP \perp AB$

The converse is also true: a line drawn through the end point of a radius and perpendicular to it is a tangent to the circle. This is how you construct a tangent at a point on the circle, just draw the radius and then draw a line perpendicular to it at that point.

Worked Example

A tangent PQ at a point P of a circle of radius 5 cm meets a line through the centre O at a point Q such that $OQ = 13$ cm. Find the length of PQ.

Since the radius OP is perpendicular to the tangent PQ at P, $\triangle OPQ$ is right-angled at P.

$$OQ^2 = OP^2 + PQ^2 \quad (\text{Pythagoras})$$

$$13^2 = 5^2 + PQ^2 \implies 169 = 25 + PQ^2 \implies PQ^2 = 144$$

So, **PQ = 12 cm.**

Theorem: Lengths of Tangents from an External Point

The second theorem says that the two tangents you can draw from a point outside the circle are not just any two lines, they are equal in length.

Theorem 10.2, The lengths of tangents drawn from an external point to a circle are equal.

If PA and PB are two tangents drawn from an external point P to a circle with centre O, touching the circle at A and B, then $PA = PB$.

Proof. Join OA, OB and OP. Since the tangent is perpendicular to the radius at the point of contact, $\angle OAP = 90^\circ$ and $\angle OBP = 90^\circ$.

In right-angled triangles OAP and OBP:

$OA = OB$ (radii of the same circle)

$OP = OP$ (common hypotenuse)

$\angle OAP = \angle OBP = 90^\circ$

So, $\triangle OAP \cong \triangle OBP$ by RHS congruence (Right angle, Hypotenuse, Side). So, by CPCT,

$PA = PB$

This congruence also tells us two extra facts that are useful in problems: (i) OP bisects $\angle APB$, that is, $\angle APO = \angle BPO$, and (ii) OP bisects $\angle AOB$, that is, $\angle AOP = \angle BOP$. The line joining the external point to the centre bisects both the angle between the tangents and the angle between the radii to the points of contact.

Worked Example

From a point P outside a circle, two tangents PA and PB are drawn. If $PA = 8\text{ cm}$, what is PB? If $\angle APB = 60^\circ$, what is $\angle AOB$?

By Theorem 10.2, the lengths of tangents from an external point are equal, so **$PB = PA = 8\text{ cm}$** .

Join OA, OB and OP. In quadrilateral OAPB, $\angle OAP = 90^\circ$ and $\angle OBP = 90^\circ$ (radius $\perp$ tangent). The sum of interior angles of a quadrilateral is 360° , so

$$\angle AOB + \angle APB + \angle OAP + \angle OBP = 360^\circ$$

$$\angle AOB + 60^\circ + 90^\circ + 90^\circ = 360^\circ \implies \angle AOB = 120^\circ$$

Alternatively, since $\triangle OAP \cong \triangle OBP$, $\angle AOP = \angle BOP$, and each is 60° , giving $\angle AOB = 120^\circ$.

A common exam application: if two tangents PA and PB are drawn from an external point P and a third tangent touches the circle at a point C on the minor arc AB and intersects PA at X and PB at Y, then the perimeter of $\triangle PXY$ equals $PA + PB = 2PA$. This follows directly by applying the theorem twice, the two tangents from X are equal ($XA = XC$) and the two from Y are equal ($YB = YC$).

Summary

- A **tangent** touches the circle at exactly one point, called the **point of contact**. A **secant** cuts the circle at two points.
- **Number of tangents from a point:** none from an interior point, one from a point on the circle, two from an exterior point.
- **Theorem 10.1:** The tangent at any point of a circle is perpendicular to the radius through the point of contact. So $\angle$ between tangent and radius $= 90^\circ$.
- The converse is also true: a line perpendicular to a radius at its end point on the circle is a tangent.
- **Theorem 10.2:** The lengths of the two tangents drawn from an external point to a circle are equal ($PA = PB$).
- The line joining the external point to the centre bisects the angle between the two tangents and the angle between the two radii to the points of contact.
- In quadrilateral OAPB formed by two radii and two tangents, $\angle AOB + \angle APB = 180^\circ$.

Practice Questions

1. A line meets a circle at exactly one point. What is this line called, and what is that point called?

2. How many tangents can be drawn to a circle from a point that is 10 cm from the centre if the radius is 6 cm? Justify your answer.

3. Prove that the tangent at any point of a circle is perpendicular to the radius through the point of contact.

4. Prove that the lengths of tangents drawn from an external point to a circle are equal.

5. Two tangents PA and PB are drawn from an external point P to a circle with centre O. If $\angle APB = 70^\circ$, find $\angle AOB$.

6. A tangent PQ at a point P of a circle of radius 6 cm meets a line through the centre O at a point Q such that $OQ = 10$ cm. Find the length of PQ.